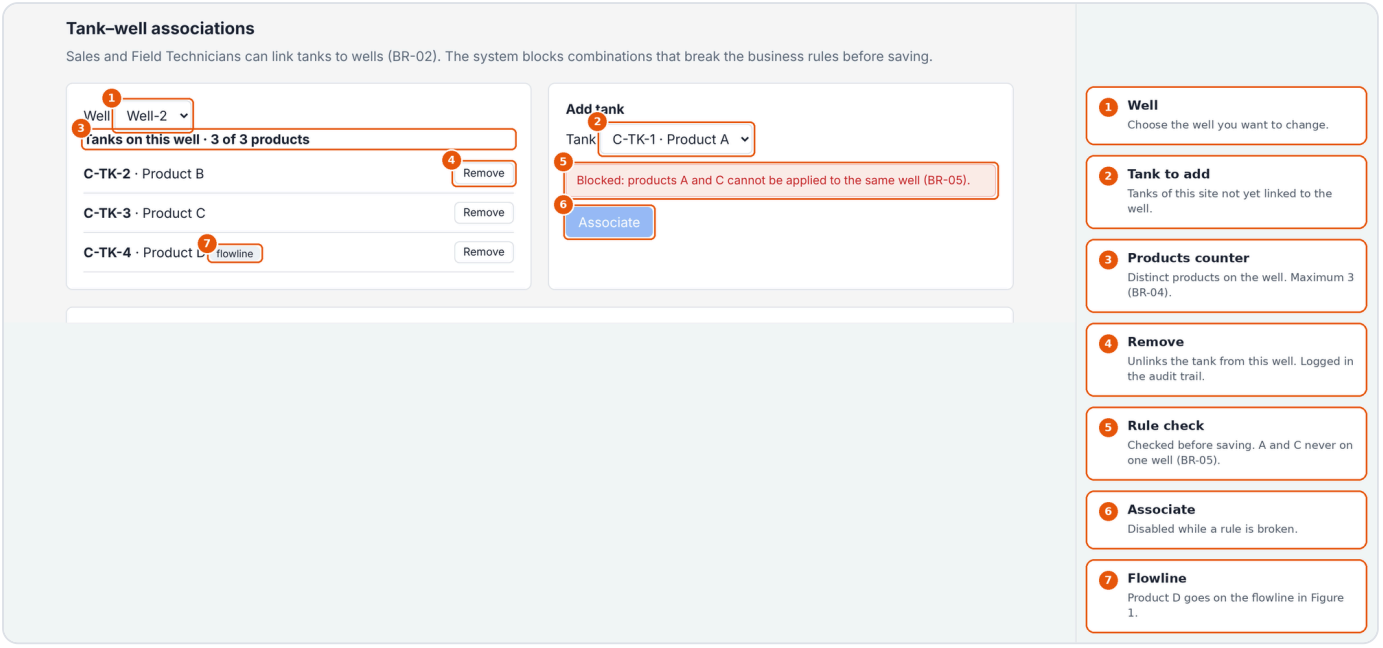
QUICK GUIDE FOR EVALUATORS

Chemical Injection Manager

How to test the prototype in 10 minutes: what each screen does, what each field means, and what to expect.



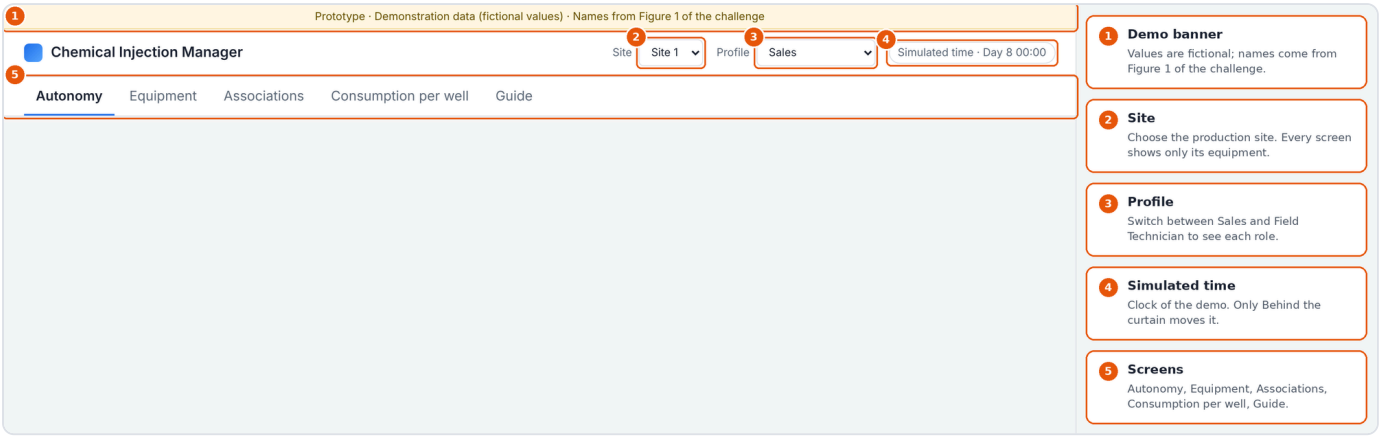
Prototype	po-struture-task-desafio-ba.vercel.app
Repository	github.com/Thiago-Sartori85/po-struture-task-desafio-ba
Project route	Miro board
Author	Thiago Sartori

1. Before you start

A **Wizard of Oz** prototype: the screens and business rules are real; the external real-time data system is played by a hidden panel called **Behind the curtain**.

Real	Simulated
Business rules, formulas, statuses, permissions by profile, validation, audit trail	Real-time data system, login, all numbers (fictional). Names come from Figure 1 of the challenge.

- **Open:** po-struture-task-desafio-ba.vercel.app in any browser. Nothing to install.
- **Simulation panel:** press **Shift + O**, click *Behind the curtain* in the footer, or add `?oz=1` to the address.
- **Start over:** reload the page or click *Reset demo* in the panel.



The header sets where you are (site), who you are (profile) and when (simulated time).

2. Test route (10 minutes)

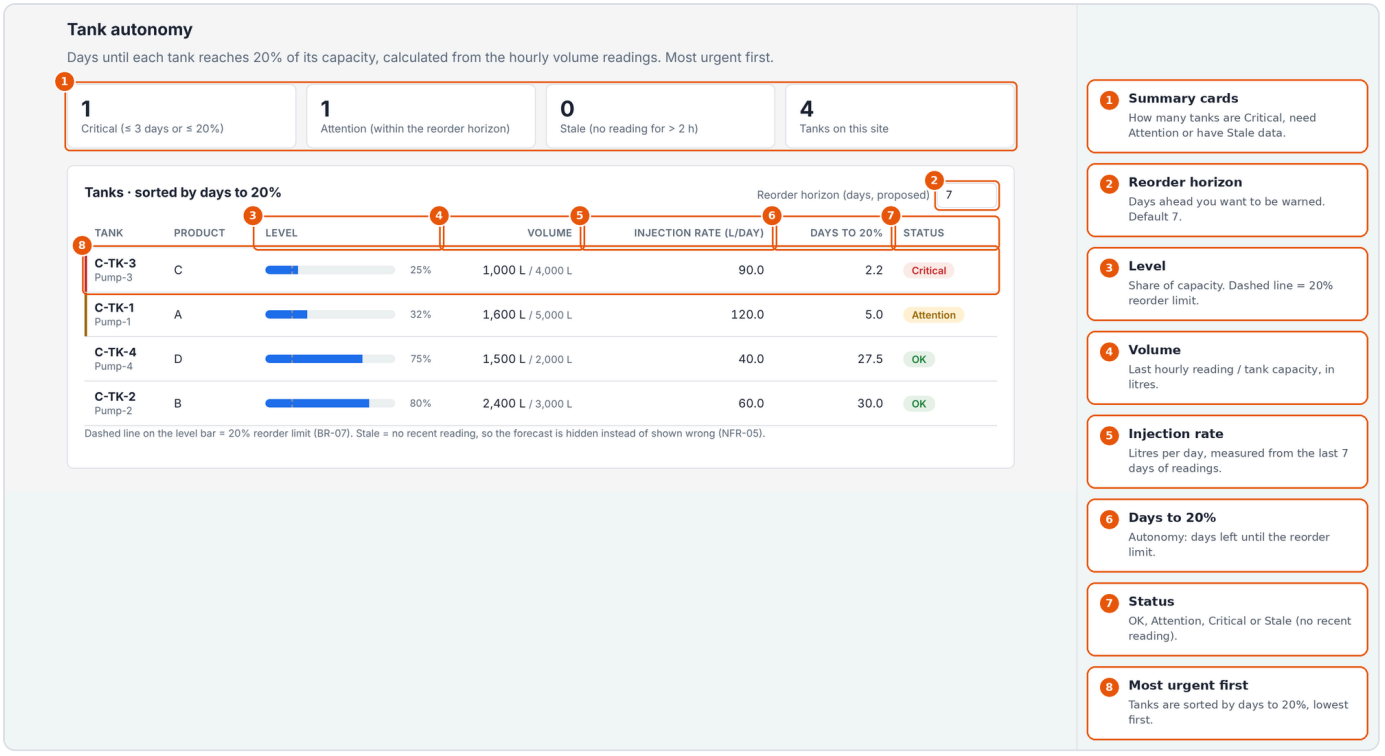
Follow the steps in order. Each one shows the expected result.

#	Do this	You should see	✓
1	Open the prototype on **Autonomy**.	C-TK-3 first, **Critical** (2.2 days). C-TK-1 **Attention** (5.0 days).	☐
2	Open **Consumption per well**.	Well-1 flagged **Conflict BR-05**: A + C, as drawn in Figure 1.	☐
3	**Associations** → Well-2 → add C-TK-1 (product A).	Red BR-05 message; *Associate* disabled.	☐
4	Choose Well-3 → add C-TK-2 (product B) → *Associate*.	Link saved, "2 of 3 products", entry in the audit trail.	☐
5	**Equipment** → click *Save tank* with empty fields.	Red validation messages under each field.	☐
6	Switch **Profile** to Field Technician.	The Equipment tab disappears (BR-01). Switch back to Sales.	☐
7	Press **Shift + O** → **+7 days**.	Clock at Day 15. C-TK-1 becomes **Critical**.	☐
8	Choose C-TK-3 → **Refill to capacity** → **+1 hour**.	C-TK-3 back to **OK**; its rate stays 90 L/day.	☐
9	Choose C-TK-2 → **Stop sensor** → +1 hour three times.	C-TK-2 **Stale**; forecast "hidden".	☐
10	Switch **Site** to Site 2.	Only TK-201 and Well-201 are shown.	☐

> The **Guide** tab inside the prototype summarizes the same rules, fields and formulas.

3. Autonomy

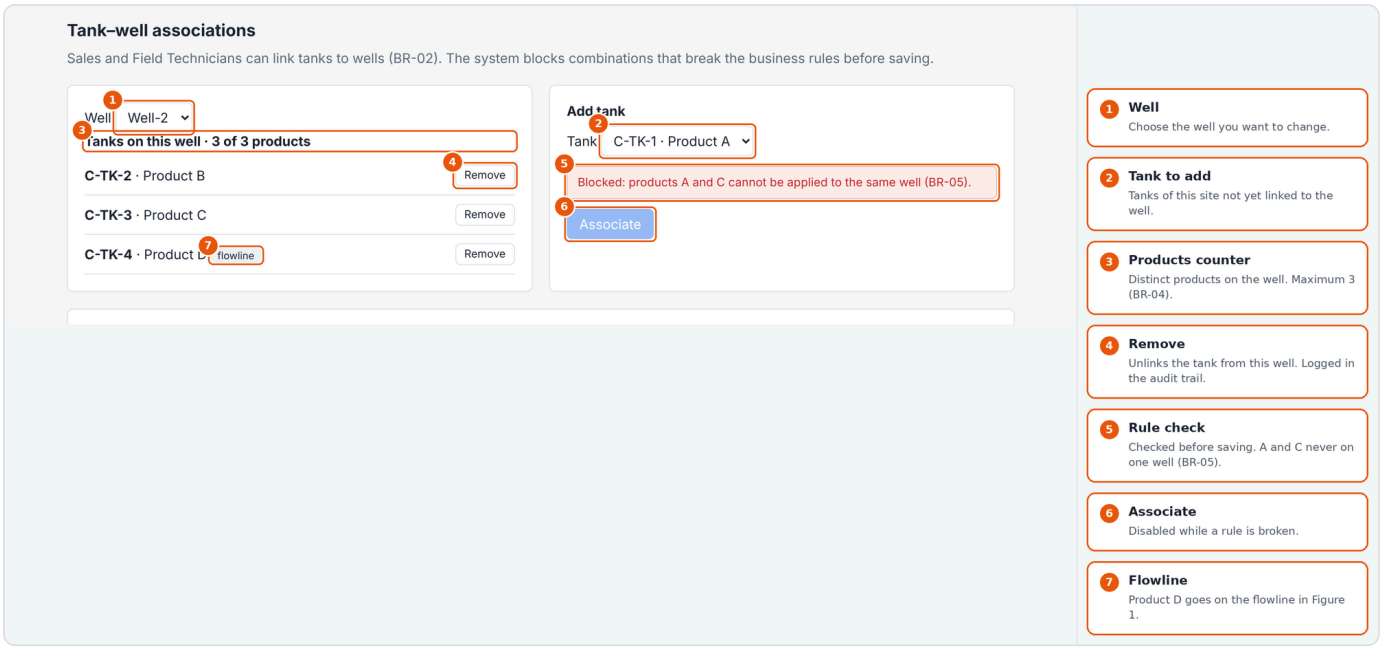
Which tanks need a delivery first? Days until each tank reaches 20% of capacity, most urgent on top.



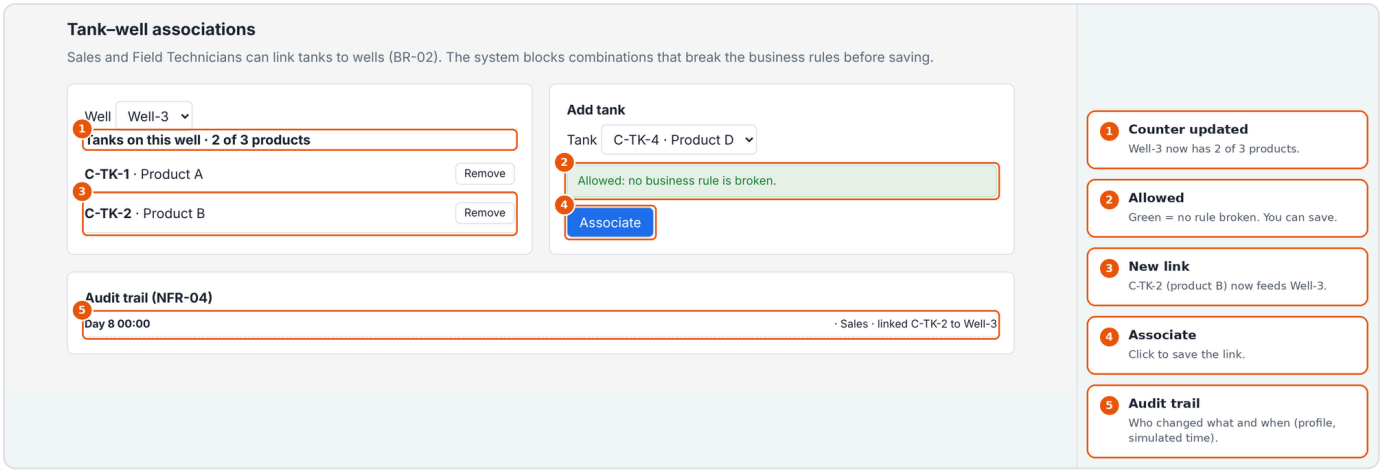
Status	Meaning
Stale	No reading for more than 2 hours; forecast hidden
Critical	At or below 20% of capacity, or ≤ 3 days
Attention	Within the reorder horizon (default 7 days)
OK	Beyond the horizon

4. Associations

Link tanks to wells. Rules are checked **before saving**; invalid links cannot be saved.



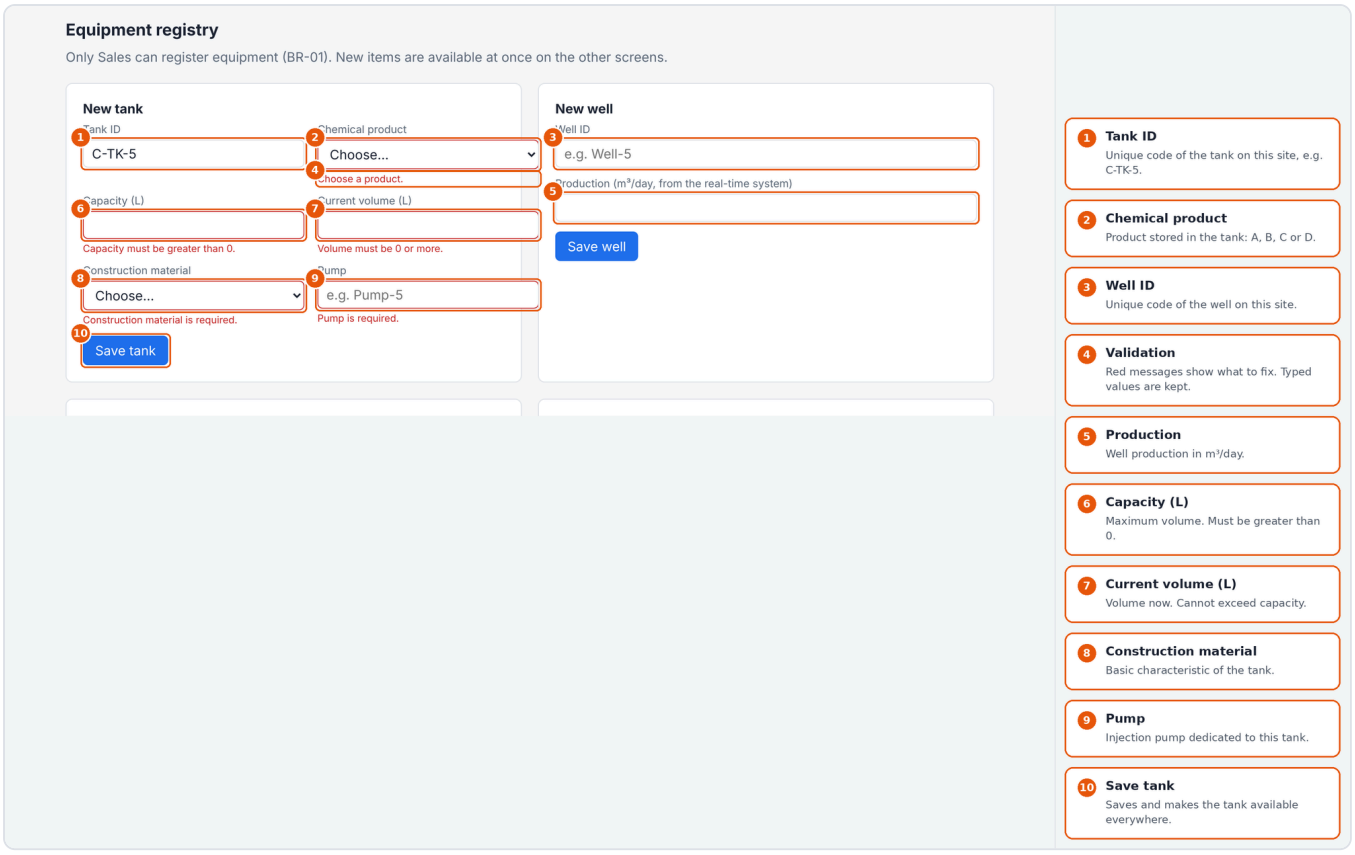
Blocked: Well-2 already has product C.

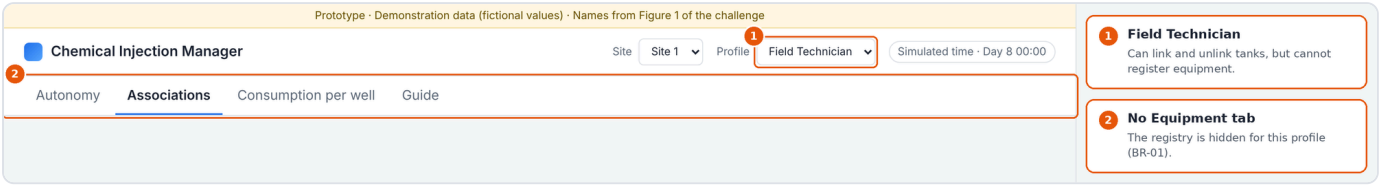


Allowed and recorded in the audit trail.

5. Equipment

Register tanks and wells. Visible to **Sales** only.

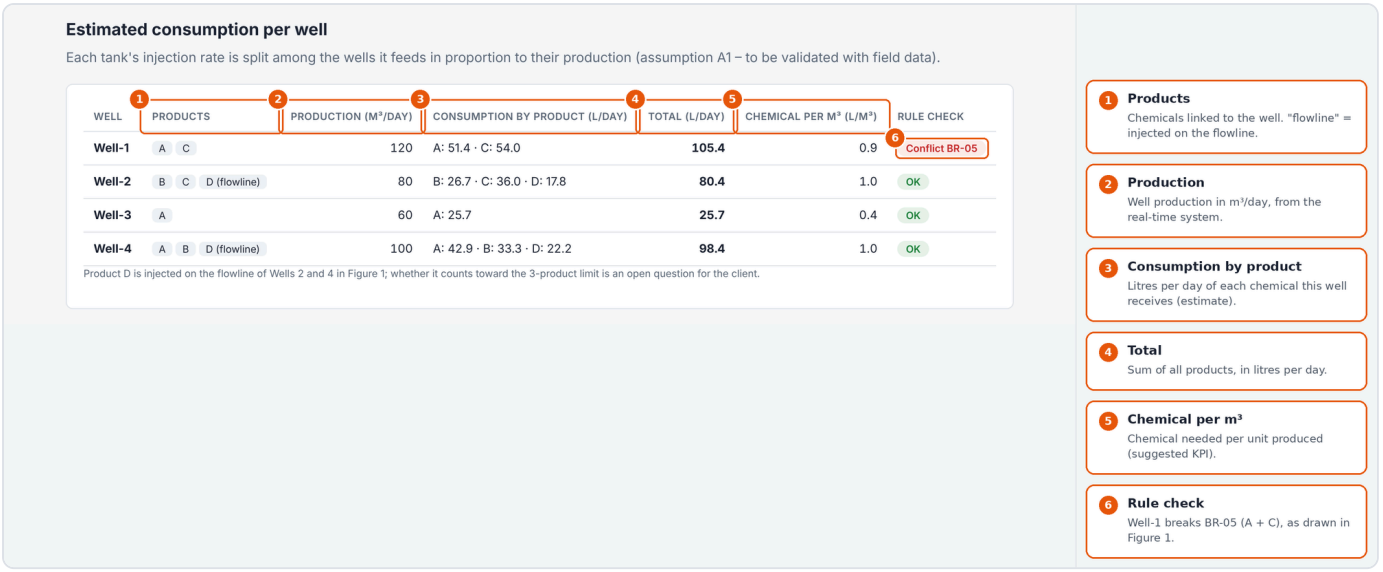




Field Technician profile: no Equipment tab.

6. Consumption per well

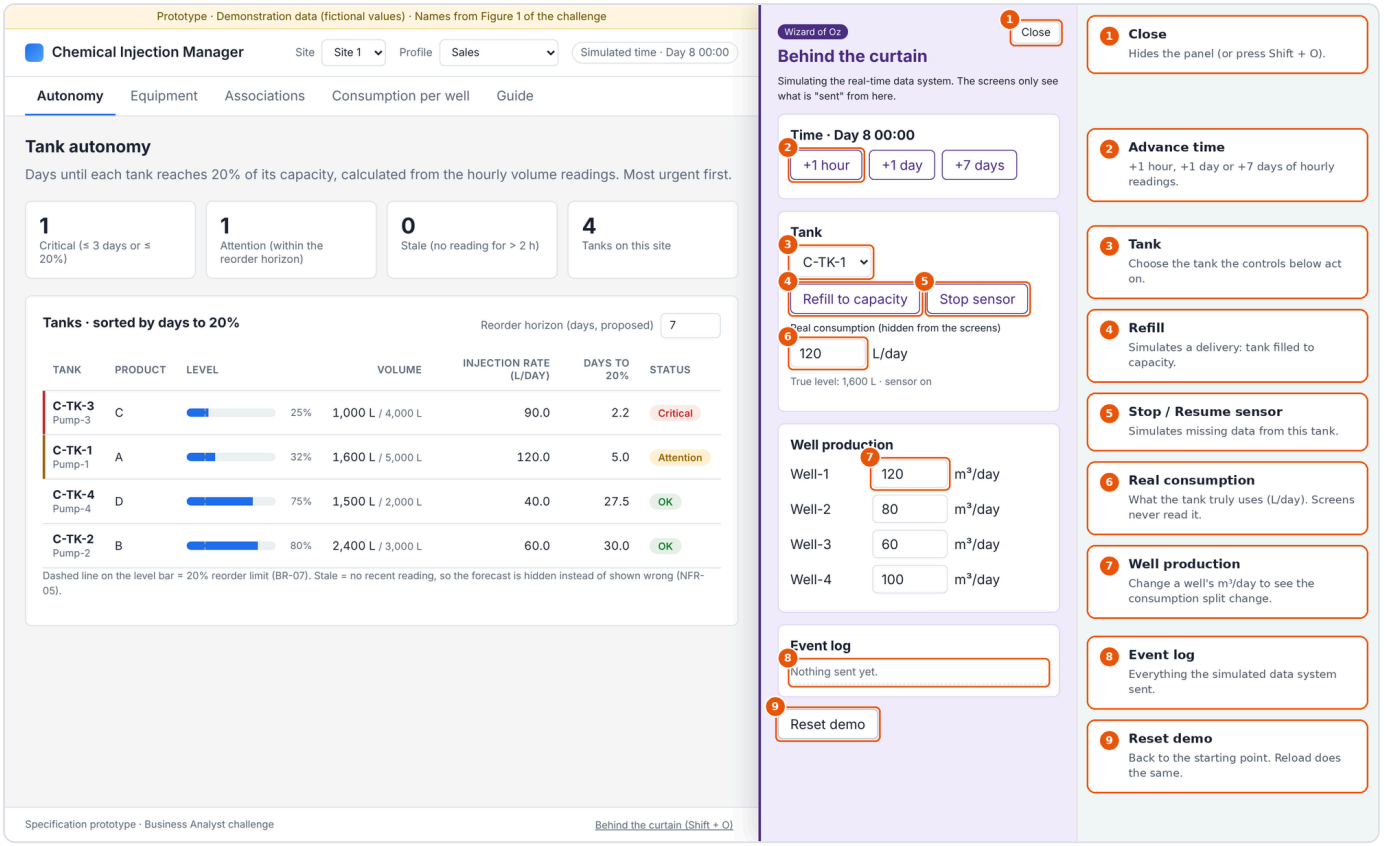
Estimated chemical each well receives per day: each tank's rate is split by well production (assumption A1).

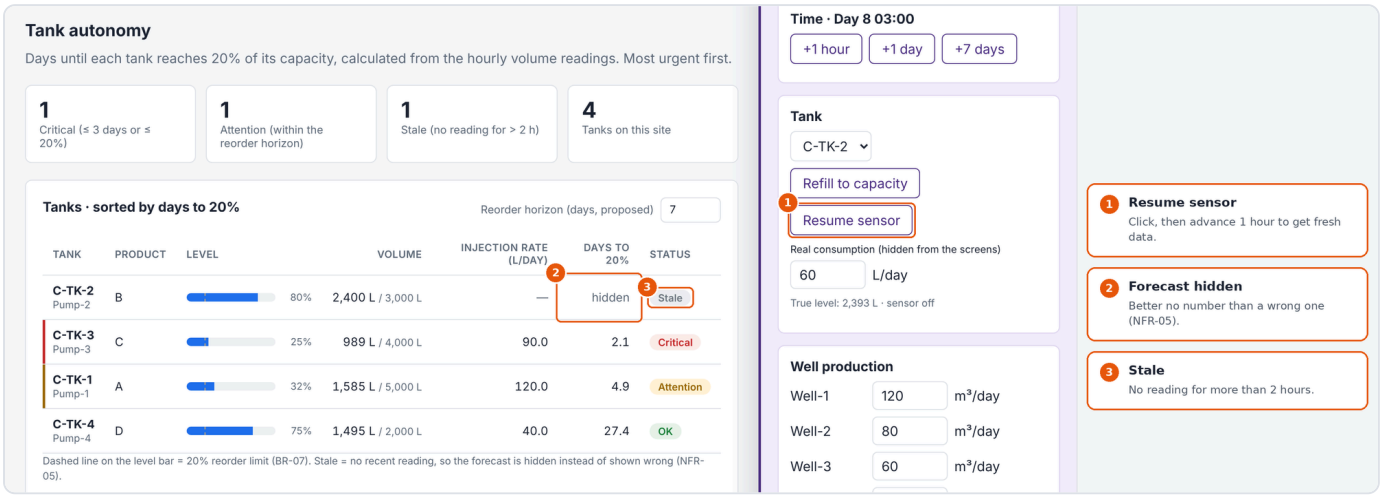



```
Injection rate (L/day) = volume drop over the last 7 days ÷ days covered  (refills excluded)
Days to 20%            = (volume − 0.20 × capacity) ÷ injection rate
Well consumption       = Σ tank rate × (well production ÷ production of all wells fed by that tank)
```

7. Behind the curtain

Plays the real-time data system. The screens never read the "real consumption" it holds: they measure it from volume readings, as the real system would.





Sensor stopped for 3 hours: the tank is Stale and its forecast is hidden.

8. Rules at a glance

Rule	What it says	Try it
BR-01	Only Sales registers equipment	Profile → Field Technician
BR-02	Sales and Field Technicians link tanks to wells	Associations, both profiles
BR-04	Maximum 3 chemical products per well	Counter "n of 3 products" in Associations. With products A–D, a 4th product always also breaks BR-05, so that message is shown; BR-04 is covered by unit tests.
BR-05	Products A and C never on the same well	Add C-TK-1 (A) to Well-2
BR-06	Equipment belongs to one site; many sites	Site selector
BR-07	Autonomy counted until 20% of capacity	Days to 20%, dashed line on the level bar

Good to know

- Data lives in the browser tab only; reloading resets the demo.
- Shift + O does nothing while typing in a field: click outside first.
- A newly registered tank shows "collecting readings" until it has consumption: set it in the panel and advance time.
- The 2-hour, 3-day and 7-day thresholds are proposed parameters to agree with the client.
- Full specification, wireframes and source code: see the repository README.